Math 596: Complex Analysis I

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Abstract

Math 596 taught by Professor Dmitry Chelkak

Contents

Chapter 1

Reminders on $\mathbb{C}$, basic facts, Cauchy-Riemann Equations

Lecture 1: Reminders on $\mathbb{C}$

1.1 Basic properties of $\mathbb{C}$

01 Sep. 14:30

Here are some basic facts about $\mathbb{C}$.

- $\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$. As a field, the multiplication is defined by $(x+iy)(s+it) = (xs - yt) + i(xt + ys)$.
- Re, Im takes real and imaginary parts of a complex number, respectively. $\mathrm{Re}(x+iy) = x$, $\mathrm{Im}(x+iy) = y$.
- $\overline{z} = x - iy$ if $z = x + iy$ (conjugation), and $\overline{z_1} + \overline{z_2} = \overline{z_1 + z_2}$, $\overline{z_1} \cdot \overline{z_2} = \overline{z_1 \cdot z_2}$.
- $|z| = \sqrt{x^2 + y^2}$, and $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$, $|z|^2 = z \cdot \overline{z}$.
- $|z_1 + z_2| \leq |z_1| + |z_2|$ (triangle inequality). This gives a metric on $\mathbb{C}$,
- $z_n \xrightarrow{n\to\infty} z$ if and only if $|z_n - z| \xrightarrow{n\to\infty} 0$.
- Exercise: $z_n \xrightarrow{n\to\infty} z$ if and only if $\mathrm{Re}(z_n) \xrightarrow{n\to\infty} \mathrm{Re}(z)$ and $\mathrm{Im}(z_n) \xrightarrow{n\to\infty} \mathrm{Im}(z)$.
- $\mathbb{C}$ is a complete metric space.
- Addition, multiplication and conjugation are continuous functions.
- Exponential functions: (bad way to define: $e^{x+iy} = e^x(\cos y + i \sin y)$, and we can check that $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$). better way:
$$e^z := \sum_{n=0}^{\infty} \frac{z^n}{n!}$$
Since $\sum_{k=0}^{n} \frac{z^k}{k!}$ is a Cauchy sequence (because $e^{|}z|$ is Cauchy), we see that e^z converges and is

well-defined. Now we check that $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$:

$$\begin{aligned} e^{z_1}e^{z_2} &= \sum_{n_1=0}^{\infty} \frac{z_1^{n_1}}{n_1!} \sum_{n_2=0}^{\infty} \frac{z_2^{n_2}}{n_2!} \\ &= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \frac{z_1^{n_1} z_2^{n_2}}{n_1! n_2!} \\ &= \sum_{n=0}^{\infty} \sum_{n_1=0}^{n} \binom{n}{n_1} \frac{z_1^{n_1} z_2^{n-n_1}}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(z_1+z_2)^n}{n!} \\ &= e^{z_1+z_2} \end{aligned}$$

where the exchange of summation is justified by the absolute convergence of the series (it is majorated by $e^{|z_1|}e^{|z_2|} < \infty$).

- $\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$, $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$.
- Polar representation of complex numbers: $z = re^{i\theta}$, where $r = |z|$ and $\theta = \arg(z)$. Assume $z \neq 0$, $r \in \overline{R_+}$ and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$.

Notation:

- $\operatorname{Arg} z \in (-\pi, \pi]$ is called the principal value of the argument of z.
- $\arg z = \{\operatorname{Arg} z + 2\pi n : n \in \mathbb{Z}\}$.
- $\arg(z_1 z_2) = \arg z_1 + \arg z_2$.
- $\arg(1/z) = \arg(\overline{z}) = -\arg z$.

Open sets in $\mathbb{C}$

Notation:

- $D_r(z) = \{z' \in \mathbb{C} : |z' - z| < r\}$.
- $\overline{D_r}(z) = \{z' \in \mathbb{C} : |z' - z| \leq r\}$.
- $C_r(z) = \partial D_r(z)\{z' \in \mathbb{C} : |z' - z| = r\}$.
- $\mathbb{D} = D_1(0) = \{z \in \mathbb{C} : |z| < 1\}$.
- $\mathbb{T} = \partial\mathbb{D} = \{z \in \mathbb{C} : |z| = 1\}$.

Definition 1.1.1. A subset $\Omega \subset \mathbb{C}$ is *open* if for any $z \in \Omega$, there exists $r > 0$ such that $D_r(z) \subset \Omega$.

Definition 1.1.2. A subset $\Omega \subset \mathbb{C}$ is called *connected* if there does not exist a pair of disjoint open sets (open in Ω), $\Omega_1, \Omega_2 \neq \varnothing$, such that $\Omega = \Omega_1 \sqcup \Omega_2$.

Definition 1.1.3. Ω is called *path-connected* if for any $z_1, z_2 \in \Omega$, there exists a continuous map $\gamma : [0, 1] \to \Omega$ such that $\gamma(0) = z_1$ and $\gamma(1) = z_2$.

Lemma 1.1.1. If $\Omega \subset \mathbb{C}$ is open, then Ω is connected if and only Ω is path-connected. Moreover, in this case γ can be taken as a polyline with finitely many vertical and horizontal segments.

Proof. "$\Leftarrow$": Let $\Omega = \Omega_0 \sqcup \Omega_1$ be a disjoint union of two open sets. Choose $z_0 \in \Omega_0$ and $z_1 \in \Omega_1$. Consider $\gamma : [0,1] \to \Omega$ such that $\gamma(0) = z_0$ and $\gamma(1) = z_1$. Then $\gamma^{-1}(\Omega_0)$ and $\gamma^{-1}(\Omega_1)$ are disjoint open sets in $[0,1]$ whose union is $[0,1]$, which is impossible.

"$\Rightarrow$": Pick $z_0 \in \Omega$. Let

$$\Omega_0 = \{z \in \Omega : z \text{ is connected to } z_0 \text{ by a polyline with finitely many vertical and horizontal segments}\}.$$

Note that Ω_0 is both open and closed. So we must have $\Omega_0 = \Omega$ since Ω is connected. ■

From now on, unless otherwise stated, Ω is assumed to be open.

1.1.1 Continuity and differentiability

Definition 1.1.4. $f : \Omega \to \mathbb{C}$ is *continuous* at $z_0 \in \Omega$ if for any U_0 such that $f(z_0) \in U_0$, there exists open V such that $z_0 \in V \subseteq \Omega$, and $f(V) \subseteq U_0$.

Remark. An equivalent definition is that for any sequence $z_0 \xrightarrow{n\to\infty} z$, we have $f(z_n) \xrightarrow{n\to\infty} f(z)$.

Problem 1.1.1. f is continuous if and only if $\operatorname{Re} f$ and $\operatorname{Im} f$ are continuous.

As previously seen. $f : \mathbb{R} \to \mathbb{R}$ is differentiable at x_0 if there is $f'(x_0) \in \mathbb{R}$ such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + o(|x - x_0|), \quad x \to x_0.$$

$f : \mathbb{R}^2 \to \mathbb{R}$ is differentiable at (x_0, y_0) if there are $a, b \in \mathbb{R}$ such that

$$u(x,y) = u(x_0,y_0) + a(x - x_0) + b(y - y_0) + o(|x - x_0| + |y - y_0|), \quad (x,y) \to (x_0,y_0).$$

Note that this is a stronger condition than merely requiring that the partial derivatives $\partial_x u$ and $\partial_y u$ exist at (x_0, y_0).

Definition 1.1.5. $f : \Omega \to \mathbb{C}$ is called $(\mathbb{C})$-*differentiable* at $z_0 \in \Omega$ if there exists $f'(z_0) \in \mathbb{C}$ such that

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + o(|z - z_0|), \quad z \to z_0.$$

Then $a = \lim_{n\to\infty} \frac{f(z)-f(z_0)}{z-z_0} := f'(z_0)$ is called the derivative of f at z_0.

Lecture 2: Differentiability and Cauchy-Riemann Equations

31 Aug. 9:00

As previously seen.

(a) $f : \mathbb{R} \to \mathbb{R}$ is differentiable at x_0 if there is $a \in \mathbb{R}$ such that

$$f(x) = f(x_0) + a(x - x_0) + o(|x - x_0|), \quad x \to x_0.$$

C^1 is denoted as the class of functions $f : \mathbb{R} \to \mathbb{R}$ such that f is differentiable and f' is continuous. There are continuous, differentiable functions but f' is not continuous. For example, $f(x) = x^2 \sin(1/x)$.

(b) $u : \mathbb{R}^2 \to \mathbb{R}$ is differentiable at (x_0, y_0) if there are $a, b \in \mathbb{R}$ such that

$$u(x,y) = u(x_0,y_0) + a(x - x_0) + b(y - y_0) + o(|x - x_0| + |y - y_0|), \quad (x,y) \to (x_0,y_0).$$

Notice here $x - x_0, y - y_0 \to a(x - x_0) + b(y - y_0)$ is a linear map. Moreover, in this case we have $a = \frac{\partial u}{\partial x}(x_0, y_0)$ and $b = \frac{\partial u}{\partial y} y(x_0, y_0)$.

Remark. Existence of $\frac{\partial u}{\partial x}(x_0, y_0)$ and $\frac{\partial u}{\partial y} y(x_0, y_0)$ does not imply differentiability. For example, $u(x, y) = \frac{xy}{x^2+y^2}$. It is not even continuous at $(0, 0)$!

Definition 1.1.6. $f : \Omega \to \mathbb{C}$ is called $(\mathbb{C})$-*differentiable* at $z_0 \in \Omega$ if there exists $a \in \mathbb{C}$ such that

$$f(z) = f(z_0) + a(z - z_0) + o(|z - z_0|), \quad z \to z_0.$$

Definition 1.1.7. f is called *holomorphic* at z_0 if there is an open set $U \subset \Omega$ such that $z_0 \in U$ and f is differentiable everywhere in U.

Remark (Warning). We did not assume continuity for f'. But this more general fact is true!

Remark. There is a notion of *analytic* functions, where f is analytic at z_0 if $f \in C^\infty$ and the Taylor series converges to f near z_0. These two notions turns out to be equivalent, which is a deep result that we are going to prove.

Note. Differentiability at z_0 does not imply holomorphicity at z_0. See HW for example.

Lemma 1.1.2. If $f, g : \Omega \to \mathbb{C}$ are holomorphic at $z_0 \in \Omega$, then $f + g$, fg, and f/g (if $g(z_0) \neq 0$) are holomorphic at z_0.

Proof. Mimic proof in real analysis. ■

Example. e^z is holomorphic everywhere, and so are $\sin z$, $\cos z$.

Proof.

$$\begin{aligned} e^z &= e^{z_0} e^{z - z_0} \\ &= e^{z_0}\left(1 + (z - z_0) + O((z - z_0)^2)\right) \end{aligned}$$

This is true since

$$\begin{aligned} \left|\sum_{n=2}^{\infty} \frac{(z - z_0)^n}{n!}\right| &\leq |z - z_0|^2 \sum_{n=0}^{\infty} \frac{|z - z_0|^n}{n + 2} \\ &\leq |z - z_0|^2 e^{|z - z_0|} \end{aligned}$$

■

Proposition 1.1.1. Let $u : \Omega \to \mathbb{R}$ be such that $u_x := \frac{\partial u}{\partial x}$ and $u_y := \frac{\partial u}{\partial y}$ exist everywhere in Ω and are continuous at (x_0, y_0). Then u is differentiable at (x_0, y_0).

Proof. We have

$$\begin{aligned} u(x, y) - u(x_0, y_0) &= (u(x, y) - u(x, y_0)) + (u(x, y_0) - u(x_0, y_0)) \\ &= u_y(x, y_1)(y - y_0) + u_x(x_1, y_0)(x - x_0) (\text{here continuity is not required}) \\ &= (u_y(x_0, y_0) + o(1))(y - y_0) + (u_x(x_0, y_0) + o(1))(x - x_0) \\ &= u_y(x_0, y_0)(y - y_0) + u_x(x_0, y_0)(x - x_0) + o(|x - x_0| + |y - y_0|) \end{aligned}$$

where y_1 is between y_0 and y, and x_1 is between x_0 and x. The second line is by the mean value theorem, and the third line is by continuity of u_x and u_y. ■

Theorem 1.1.1 (Cauchy-Riemann equations). $f = u + iv : \Omega \to \mathbb{C}$ is $\mathbb{C}$-differentiable at $z_0 = x_0 + iy_0$ if and only if u, v are differentiable at (x_0, y_0), and

$$\begin{aligned} u_x(x_0, y_0) &= v_y(x_0, y_0) \\ u_y(x_0, y_0) &= -v_x(x_0, y_0) \end{aligned}$$

Proof. "⇒": If $f'(z_0) = a + ib$, then $u(z) = \operatorname{Re}(f(z_0) + f'(z_0)(z - z_0) + o(|z - z_0|))$. Then

$$f'(z_0)(z - z_0) = (a + ib)((x - x_0) + i(y - y_0)) = (a(x - x_0) - b(y - y_0)) + i(b(x - x_0) + a(y - y_0))$$

Then we see that

$$u_x(x_0, y_0) = a, \quad u_y(x_0, y_0) = -b, \quad v_x(x_0, y_0) = b, \quad v_y(x_0, y_0) = a$$

Conversely, if we denote $a = u_x(x_0, y_0) = v_y(x_0, y_0)$ and $b = v_x(x_0, y_0) = -u_y(x_0, y_0)$, then, since u, v are differentiable, we have

$$\begin{aligned} u(x, y) &= u(x_0, y_0) + a(x - x_0) - b(y - y_0) + o(|x - x_0| + |y - y_0|) \\ v(x, y) &= v(x_0, y_0) + b(x - x_0) + a(y - y_0) + o(|x - x_0| + |y - y_0|) \end{aligned}$$

Then we have

$$f(z) - f(z_0) = (u(x, y) - u(x_0, y_0)) + i(v(x, y) - v(x_0, y_0)) = (a + ib)(z - z_0) + o(|z - z_0|).$$

■

1.2 Logarithm

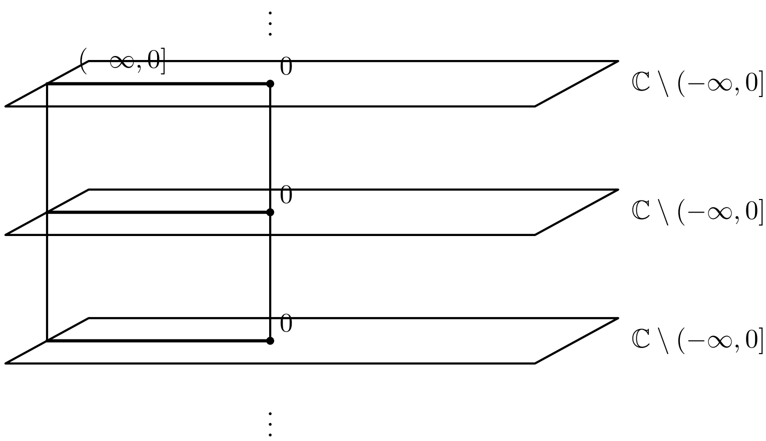
Roughly speaking, $w = \log z$ is a solution to the equation $e^w = z$. Then we get $w = \log|z| + i \operatorname{Arg} z$, as $e^{u+iv} = e^u e^{iv} = e^u(\cos v + i \sin v)$. We first need $z \neq 0$. But there is another problem: continuity. $\operatorname{Arg} z$ is not a continuous function.

There are two ways to resolve it:

The first way is to define another function $\operatorname{Log} z = \log|z| + i \operatorname{Arg}(z)$. Then we define $\log z = \{\operatorname{Log} z + 2\pi i n \mid n \in \mathbb{Z}\}$, which is a multi-valued function.

The second way, going back to the definition of $\operatorname{Log}(z)$, is to define it on $\mathbb{C} \setminus (-\infty, 0]$.

1.2.1 Idea of a Riemann surface



The idea of a Riemann surface is to have infinitely many sheets of $\mathbb{C}\setminus(-\infty,0]$ and glue them together along the branch cut $(-\infty,0]$. We can formally define it as

$$\mathbb{S}=\{(r,\theta):r\in\mathbb{R}^+,\theta\in\mathbb{R}\}$$

One easily get a projection $\pi:\mathbb{S}\to\mathbb{C}\setminus\{0\}$ by $(r,\theta)\mapsto re^{i\theta}$. Moreover, we can define $\log:\mathbb{S}\to\mathbb{C}$ by $(r,\theta)\mapsto\log r+i\theta$. This is a continuous function, and it is a single-valued function.

We have the following commutative diagram:

$$\begin{tikzcd}
\mathbb{S} \arrow[d, "\pi"] \arrow[dr, "\log"] & \\
\mathbb{C}\setminus\{0\} \arrow[r, "\mathrm{Log}"] & \mathbb{C}
\end{tikzcd}$$

Remark. Tornado ends our class...

Appendix